

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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BEST GNSS WINDOW Sun, Jul 12 · 2:00 PM – 6:00 PM | PDOP 0.97 · 35 sv | NEAREST CORS
PUM · 2.9 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
PUM · 2.9 mi

GENERATED
-

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	66 ft	13.0° E	X
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.6

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ⚠ DEGRADED

16.5 TECU

-8.9 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.6). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Sun, Jul 12	0.97	2:00 PM – 6:00 PM	35	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 12, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window
PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Sunday, Jul 12

WEEKEND

Overcast

11 mph

GO

High 88°F Low 56°F Wind max 11 mph Gusts 13 mph Precip 0.00" Sunrise 5:56 AM Sunset 8:33 PM

Waning Crescent 6%

GNSS: Avg PDOP 0.97 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY						
	PDOP			SATS		RATING
6:00 AM	0.87			36 sv		IDEAL
7:00 AM	0.91			34 sv		IDEAL
8:00 AM	1.06			29 sv		GOOD
9:00 AM	1.00			33 sv		GOOD
10:00 AM	1.00			29 sv		GOOD
11:00 AM	1.06			27 sv		GOOD
12:00 PM	0.96			29 sv		IDEAL
1:00 PM	1.00			28 sv		GOOD
2:00 PM	0.90			34 sv		IDEAL
3:00 PM	0.98			33 sv		IDEAL
4:00 PM	0.98			34 sv		IDEAL
5:00 PM	0.86			38 sv		IDEAL
6:00 PM	1.01			32 sv		GOOD
7:00 PM	1.04			30 sv		GOOD



GO — GOOD CONDITIONS

Partly cloudy | Wind: 10.6 mph (gusts 12.5 mph) | Precip: 0" | Temp: 56.3°–88.4°F



DRONE WIND ANALYSIS



GO Peak: 10.6/12.5 mph S

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	3/5 mph		SSE
6 AM	4/7 mph		ESE
7 AM	2/3 mph		ESE
8 AM	3/3 mph		W
9 AM	3/4 mph		SW
10 AM	5/5 mph		SSW
11 AM	7/9 mph		S
12 PM	7/8 mph		S
1 PM	6/7 mph		S
2 PM	7/9 mph		S
3 PM	8/10 mph		S
4 PM	11/13 mph		SSW
5 PM	10/12 mph		SSW
6 PM	8/11 mph		SSW
7 PM	8/11 mph		S
8 PM	7/11 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

Job: N/A | Site: N/A | Location: 38.3459, -122.3047 (Napa County, CA)
Report Generated: Sunday, July 12, 2026 | Survey Start: Sunday, July 12, 2026

VERDICT: GO

Clear favorable PDOP geometry (0.93–0.97 across the day) combined with acceptable weather and moderate ionospheric conditions support survey execution. No critical obstacles present.

BEST SURVEY WINDOW

2:00 PM – 6:00 PM local time (Sunday, July 12, 2026)
Reason: Peak satellite geometry with PDOP 0.93 and 35-satellite constellation (GPS+GLONASS+Galileo+BeiDou). Window coincides with afternoon stability and lowest DOP values of the day.

BEST 4-HOUR STATIC GNSS SESSION

2:00 PM – 6:00 PM (as above)
Occupation time: 2 hours minimum per baseline (4-hour preferred given moderate Kp 4.7 and TEC depression of –8.9 TECU). Multi-constellation tracking will mitigate ionospheric delay. Use 13° elevation mask per terrain analysis. Verify PRN 11 auto-exclusion in receiver configuration.

DRONE OPERATIONS ASSESSMENT

Approved subject to local airspace coordination. Wind max 11 mph is well within tolerance for fixed-wing and multirotor platforms. Clear sky window 2:00 PM – 6:00 PM optimal for visual operations. No airspace conflicts detected. Conduct standard pre-flight checks; no weather-based restrictions.

FIELD SAFETY WARNINGS

- Moderate geomagnetic activity (Kp 4.7): Longer static sessions recommended; do not rely on rapid-occupation kinematic for cm-level accuracy.
- GPS PRN 11 unhealthy (no return estimate): Verify receiver firmware excludes this satellite automatically; manual blacklisting may be required on legacy equipment.
- Temperature range 56–88°F with overcast: Standard hydration and sun protection adequate. No heat or cold stress anticipated.
- No wildfire threat detected in reporting range; air quality acceptable (AQI 45).

SUMMARY

Sunday, July 12, 2026 presents favorable survey conditions for GNSS work in Napa County. Dominant PDOP 0.97 and multi-constellation availability ensure robust geometry; moderate ionospheric activity and clear weather favor a 2:00 PM–6:00 PM session window with extended static occupancy (2–4 hours per baseline). Site terrain imposes a 13° elevation mask with low multipath risk. One GPS satellite (PRN 11) remains unhealthy but will be auto-excluded. Proceed with planned survey using recommended occupation times and multi-constellation configuration.

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
 ↗	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

Copy All

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	697.01 ft (-9 ft)	706 ft	36K af	68%
Folsom Lake	66.5 mi	448.64 ft (-17 ft)	466 ft	792K af	81%
Lake Oroville	93.4 mi	862.11 ft (-38 ft)	900 ft	2,871K af	81%
New Melones Lake	100.1 mi	1,020.25 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	55%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	697 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457.1 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Coastal Ranges aquifer (N100CACSTL) is a fractured and intergranular system developed in Tertiary sedimentary and volcanic rocks that accumulated in marine and fluvial basins 30–50 million years ago. Recharge is primarily from local precipitation and lateral flow from upland areas, with moderate to variable availability depending on fracture connectivity and seasonal precipitation patterns. At this site, water levels range from 10–88 feet below ground surface, indicating a moderately productive aquifer suitable for domestic to light agricultural use.

FORMATION

Name	Napa Valley Tertiary Sedimentary and Volcanic Aquifer Complex
Age / Rock	Eocene to Miocene (30–50 million years old); sandstone, siltstone, shale, and volcanic flows
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on local outcrops and hillslopes; lateral flow from Vaca and Mayacamas uplands
Transit Time	5–20 years to 100 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 7.0–7.8 (est. USGS/DWR)
Hardness	est. 150–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)

Notes	No sample data available for this site. Typical for formation; moderate hardness and salinity common in Napa Valley aquifers.
TREND	
Trend	Mixed trend: 1977 readings (10–54 ft) suggest higher water levels in autumn; 2014 and 2022 readings (22–88 ft) indicate deeper levels. Long-term decline likely reflects regional groundwater development and drought.
Source	Site observations 1977–2022; est. USGS/DWR regional trend analysis
Seasonal Swing	10–30 ft typical seasonal swing (est. USGS/DWR)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The North Coast aquifer system (N100CACSTL) in this region is characterized by well-drained soils and is not located within a critically overdrafted groundwater basin, indicating sustainable water availability and minimal historical groundwater depletion. The geology of northern California's coastal areas typically consists of consolidated bedrock and younger alluvial deposits with good drainage properties, which naturally resist the clay compaction mechanism that drives subsidence in over-pumped basins. For surveyors operating in this area, this means historical elevation benchmarks are likely stable and reliable for control purposes without subsidence corrections.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain with sustainable groundwater conditions
Annual Rate	<0.5 mm/yr (est.)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA critically overdrafted basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence expected
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN	
This marine climate is mild year-round with minimal heating needs but consistent coastal wind and moisture, so your envelope should prioritize air sealing and durable weather barriers over heavy insulation. Orient primary glazing south and west to capture solar heat in winter, but protect it aggressively from wind exposure—consider recessing windows or adding wind breaks—and limit glazing on the north and east facades where cold coastal winds create thermal losses. For passive design, leverage the stable temperatures by using thermal mass (concrete or masonry) on south-facing interior walls to moderate the mild swings between day and night, which will reduce mechanical conditioning loads year-round.	
🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This coastal marine site presents a well-drained gravelly loam foundation with anticipated moderate wind exposure and low snow loads. Seismic parameters are not currently defined; consult a licensed engineer to obtain site-specific seismic design coefficients and confirm appropriate detailing. Wind design at 92 mph (Risk Category II, Exposure C) may warrant standard lateral bracing and connection design.	
FOUNDATION	
Type	Conventional spread footing anticipated pending geotechnical investigation
Basis	Well-drained gravelly loam (Coombs series) and absence of subsidence suggest spread footing viability; however, consult a licensed engineer to confirm bearing capacity and settlement potential through site-specific soil testing
STRUCTURAL IMPLICATIONS	
Seismic	Seismic design coefficients (SDC, SDS, SD1) are not currently available; consult a licensed engineer to obtain USGS or equivalent seismic hazard data for this location to determine appropriate structural detailing and foundation design

Wind Exposure	Exposure C (open terrain with scattered obstructions) suggested per ASCE 7-22 Risk Category II; site reconnaissance recommended to confirm exposure classification
Wind	Standard wind load design anticipated at 92 mph 3-second gust; no hurricane or extreme coastal wind exposure anticipated for this location
Snow	Modest ground snow load of 5.8 psf anticipated; standard roof design and drainage likely adequate, with no unusual snow drift or load combination concerns

INVESTIGATIONS & INSPECTIONS

Geotech Required	Yes — geotechnical investigation recommended to confirm soil bearing capacity, settlement characteristics, and to obtain site-specific seismic parameters currently marked as null
Liquefaction Screen	Liquefaction screening recommended once seismic design parameters are confirmed; gravelly loam and well-drained conditions may suggest lower risk, but groundwater conditions should be verified
Special Inspections	Likely to be determined pending final seismic classification; consult engineer once SDC is established
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning purposes. A licensed geotechnical engineer should evaluate soil profile (boring/testing) to confirm Site Class and subsurface conditions per ASCE 7-22 Section 20.3.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	SERIES: Coombs
DRAINAGE CLASS: Well drained	HYDRO GROUP: C
SOIL ORDER: Alfisols	SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in alluvial and colluvial materials on gently rolling terrain in the Coast Ranges and interior valleys of California. The soil profile exhibits a gravelly loam surface layer overlying a clay-enriched argillic B horizon, typical of Mediterranean climates with dry summers. Site visitors should expect moderate gravel content throughout, good trafficability on slopes under 5%, and stable excavation conditions suitable for most development activities.

Alluvial and colluvial parent materials; foothill and valley landscape positions; annual grassland and oak savanna vegetation.

DRAINAGE

Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None

ENGINEERING

Bearing Capacity	3,000–4,500 psf (shallow foundations)
Rating	Good
Shrink-Swell	Moderate — clay argillic B horizon present
Frost Action	Low
Cut Suitability	Good (gravelly material excavates readily)
Fill Suitability	Fair (clay B horizon — moisture control and compaction essential)

SEPTIC

Suitability	Conditional
Notes	Perc test required — clay loam to clay substratum limits infiltration; mounding or alternative systems may be necessary

HAZARDS

Excavation	Low to Moderate — 20–35% gravel content manageable with standard equipment
Expansive Clay	Low — clay content does not typically exceed expansion thresholds
Erosion	Low on 2–5% slopes under vegetation; moderate risk in disturbed areas lacking cover
Other	None

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sddataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 12, 2026, 7:17 PM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil well 7.2 miles away, which is well beyond any actionable distance threshold. You should verify surface title and mineral rights ownership as part of standard due diligence, but subsurface oil and gas activity does not present a constraint to site planning.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200

total in search area

10

within 5 mi

108

within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active mineral producers operate within 5 miles of the site. Basalt Rock Co (2 mi) produces clay and operates an occurrence for diatomite at 3.1 mi, while Silva Pumice Quarry (5 mi) and Mt George Pumice Deposit (3.1 mi) are active pumice producers; expect heavy truck traffic on access roads, dust generation from extraction activities, and potential noise from blasting or crushing operations during their operating hours. No historical deposits warrant Phase 1 ESA consideration based on proximity.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6

Endangered

6

Threatened

12

Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Planning Assessment ****Section 7 Consultation Triggers:**** Federal permit or funding requirements will mandate ESA Section 7 consultation, particularly if the project involves water resources, wetland permitting, or impacts to vernal pool habitat where six listed species co-occur within critical habitat. ****Most Planning-Relevant Species:**** California tiger salamander, vernal pool tadpole shrimp, and California red-legged frog present the highest constraints due to their dependence on seasonal wetlands and vernal pools common to this region; Delta smelt should be considered if project involves water diversions or discharge. Northern spotted owl and Alameda whipsnake require habitat connectivity assessment if the site includes oak woodland or grassland. ****Recommended Next Step:**** Conduct a Phase 1 biological resource survey to map jurisdictional waters, seasonal wetlands, and vegetation communities; concurrently initiate early coordination with USFWS and CDFW to identify specific survey protocols and minimize project redesign delays during formal permitting. ****For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.****

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a ****mixed oak woodland and chaparral community**** typical of the inner Coast Ranges, with blue oak and California black oak as structural dominants alongside drought-tolerant shrubs and forbs like California buckeye and canyon liveforever. The flora indicates a ****transitional foothill ecosystem**** with seasonal water availability, where shade-tolerant understory plants (California bay, California Dutchman's pipe) coexist with spring ephemerals and drought-deciduous species adapted to Mediterranean climates. The presence of moisture-indicator species like Braun's horsetail and clay mariposa lily alongside drought-adapted oaks suggests ****well-drained upland soils with intermittent seasonal seepage****—meaning grading and drainage modifications should account for possible subsurface water movement during wet months while accepting that surface drainage will be rapid during dry periods. Any site work should preserve the oak canopy and avoid compacting soils, as these native communities depend on maintained permeability and the mycorrhizal networks supporting established trees. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
<i>Crotalus oreganus oreganus</i>	635 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
<i>Latrodectus hesperus</i>	154 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
<i>Dasymutilla aureola</i>	91 obs
<i>Dasymutilla californica</i>	91 obs
<i>Dasymutilla sackenii</i>	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	1100 obs
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Area ****Highest-Priority Hazards:**** Your greatest risks are toxic plants—particularly poison oak (*Toxicodendron diversilobum*) with 1,439 observations—and Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*), which are common and potentially fatal. Black widow spiders and various stinging wasps pose secondary but serious threats. ****Practical Avoidance:**** Wear long pants, closed-toe boots, and consider gaiters; learn to identify poison oak's three-leaflet clusters and avoid touching vegetation unnecessarily. Stay alert in rocky outcrops and grassland edges where rattlesnakes shelter—watch where you step and place your hands, and never hike alone. Shake out gear and clothing before use, and wash exposed skin immediately after fieldwork. If bitten by a snake, immobilize the limb, remove constrictive items, and seek immediate medical care. ****Seasonal Note:**** Rattlesnake activity peaks March through October; poison oak remains a hazard year-round but causes more exposure in warmer months when you're in lighter clothing. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

20091665146
Total ObservationsTotal SpeciesBirdsMammalsReptilesAmphibians

BIRDS (66 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	9
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melozone crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Western Screech-Owl	<i>Megascops kennicottii</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Anna's Hummingbird	<i>Calypte anna</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3
+ 51 additional species		

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	9
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	8
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2

COMMON NAME	Scientific Name	OBS
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Rough-skinned Newt	<i>Taricha granulosa</i>	1
Western Toad	<i>Anaxyrus boreas</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site, located in the North Bay region of California, exhibits characteristics of Mediterranean oak woodland and grassland habitat with proximate aquatic features, as evidenced by the diverse assemblage of riparian and wetland-dependent species including egrets, pond turtles, and multiple frog species alongside upland generalists. Several species present warrant regulatory consideration: the Foothill Yellow-legged Frog (California Species of Special Concern), Northwestern Pond Turtle (California Species of Special Concern), and California Giant Salamander are indicators of perennial or seasonal freshwater habitat requiring protection under California Environmental Quality Act review. The reptile and amphibian diversity, particularly the high species richness of aquatic and semi-aquatic species, suggests proximity to functional riparian corridors, seasonal wetlands, or springs, while the presence of raptors and ground-nesting birds indicates relatively intact grassland/scrub habitat; moderate observation frequency suggests moderate disturbance levels compatible with rural or semi-developed land use. This wildlife community is consistent with foothill or valley transition zones with intact connectivity to water features and undeveloped open space, habitat conditions that should be evaluated for potential impacts from proposed development activities. For planning purposes only. Consult a qualified biologist for site-specific biological assessments.

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↔ excellent
7/12	30.3°	98.8%	↔ excellent
8/12	33.7°	99.3%	↔ excellent
9/12	36.9°	99.8%	↔ annual optimal
10/12	39.8°	99.8%	↔ excellent
12/12	45°	99%	↔ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Solar & Energy Planning for Your Home Your location gets about 6.6 hours of useful sunlight per day on average, which is solid for solar panels—think of it like getting a good workout most days, though winter will be lighter. Your roof's angle and color are surprisingly powerful: a dark roof will cook in the summer sun like an oven (absorbing 301 BTU per hour per square foot), so if you go with a light-colored roof, you'll cut that heat gain by two-thirds and keep your cooling bills much lower—and at your latitude, a roof pitch of about 9-in-12 is nearly perfect for capturing solar panels. The tricky part is your west-facing windows: afternoon sun pours through them with nearly triple the intensity of your south glass, turning that side of your home into a heat trap in summer, so you'll want good shading (awnings, trees, or smart glass) on that side. Here's where nature helps—a large deciduous tree on your south side is like having a free umbrella that only opens when you don't need the sun, giving you 4 hours of shade in summer while letting warming sunlight through in winter when the leaves are gone. Thermal mass—think of it as your home's ability to absorb heat during the day and release it slowly at night—depends on what your walls are made of; thicker, denser walls hold heat longer and smooth out temperature swings. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing ****38.34586°N, 122.30466°W | Napa County, CA****

****Structural Design Drivers & Climate Envelope****

IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days eliminates frost heave as a structural design driver; no frost depth means footing design is not governed by seasonal ground movement. Wind and snow loading data are not available for this location, which creates a gap in the preliminary design basis. Absent these datasets, seismic design will likely govern the structural system, as the site lies in California and standard practice assumes seismic demands control in the absence of overriding wind or snow conditions. Do not proceed with final structural framing assumptions until wind speed and snow load parameters are obtained from the latest ASCE 7 maps or site-specific wind study data.

****Geotechnical Foundation Framework****

The site is underlain by Coombs gravelly loam, 2 to 5 percent slopes, classified as Hydrologic Group C with well-drained characteristics. This soil type is suitable for shallow spread footings; the well-drained classification reduces concern for saturation-induced settlement or bearing capacity loss under normal conditions. Texture data are incomplete in the provided record, so obtain a full soil boring (minimum two locations, 15–20 feet depth) to confirm grain-size distribution, SPT N-values, and confirmation of the drainage classification under site-specific groundwater conditions. The 2 to 5 percent slope suggests modest topographic relief; assess cut-and-fill geometry and potential for lateral soil movement before footing design.

****Hazard Assessment****

No Quaternary faults lie within the search radius, and no active subsidence basin data are recorded. Oil and gas wells within ~10 miles are inactive (nearest well at 7.2 miles is plugged), eliminating subsurface fluid withdrawal as a subsidence concern. Seismic hazard remains the primary geologic risk in Napa County; retain a seismic hazard evaluation to establish probabilistic ground motion parameters (PGA, spectral accelerations) required for building code compliance. Wildfire and flood risk data were not provided; coordinate with Napa County hazard maps to close this gap if the site is in a designated fire or flood zone.

****Mining & Aggregate Context****

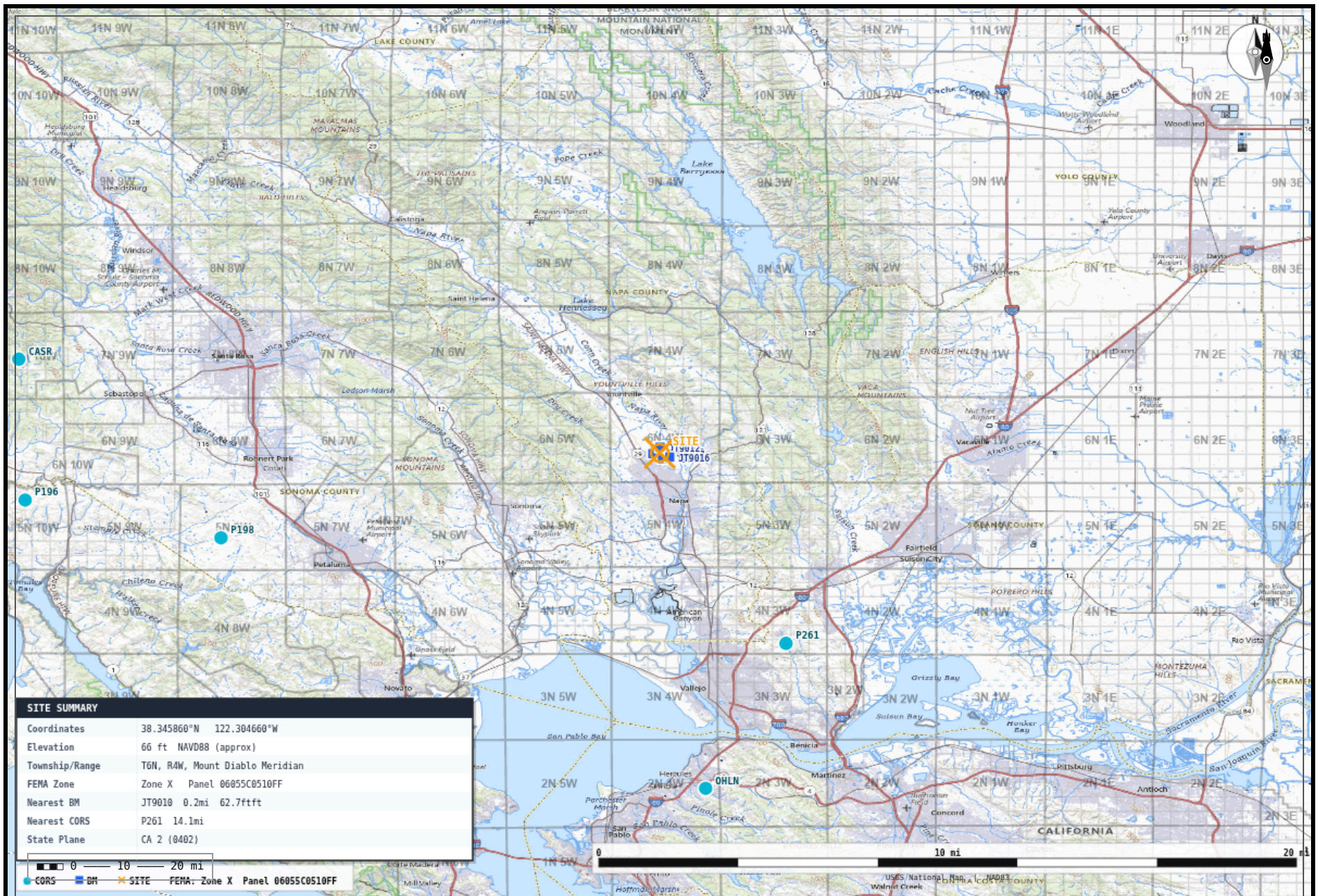
Three active extractive operations exist within a 5-mile radius: Basalt Rock Co (clay, 2 miles), Mt George Pumice Deposit (pumice, 3.1 miles), and Silva Pumice Quarry (pumice, 5 miles). The proximity of these operations warrants confirmation that no dewatering, blasting, or truck traffic from these sites will impact building performance or access. The broader 35-mile radius includes 200 MRDS mineral resource deposits; however, only the three active producers pose operational risk. Noise and dust from the nearest clay operation (2 miles) should be evaluated if the project involves sensitive use classes.

****Bottom Line****

Obtain site-specific soil boring data and seismic design parameters (USGS hazard curves or site response analysis) before finalizing foundation type and structural framing.


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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

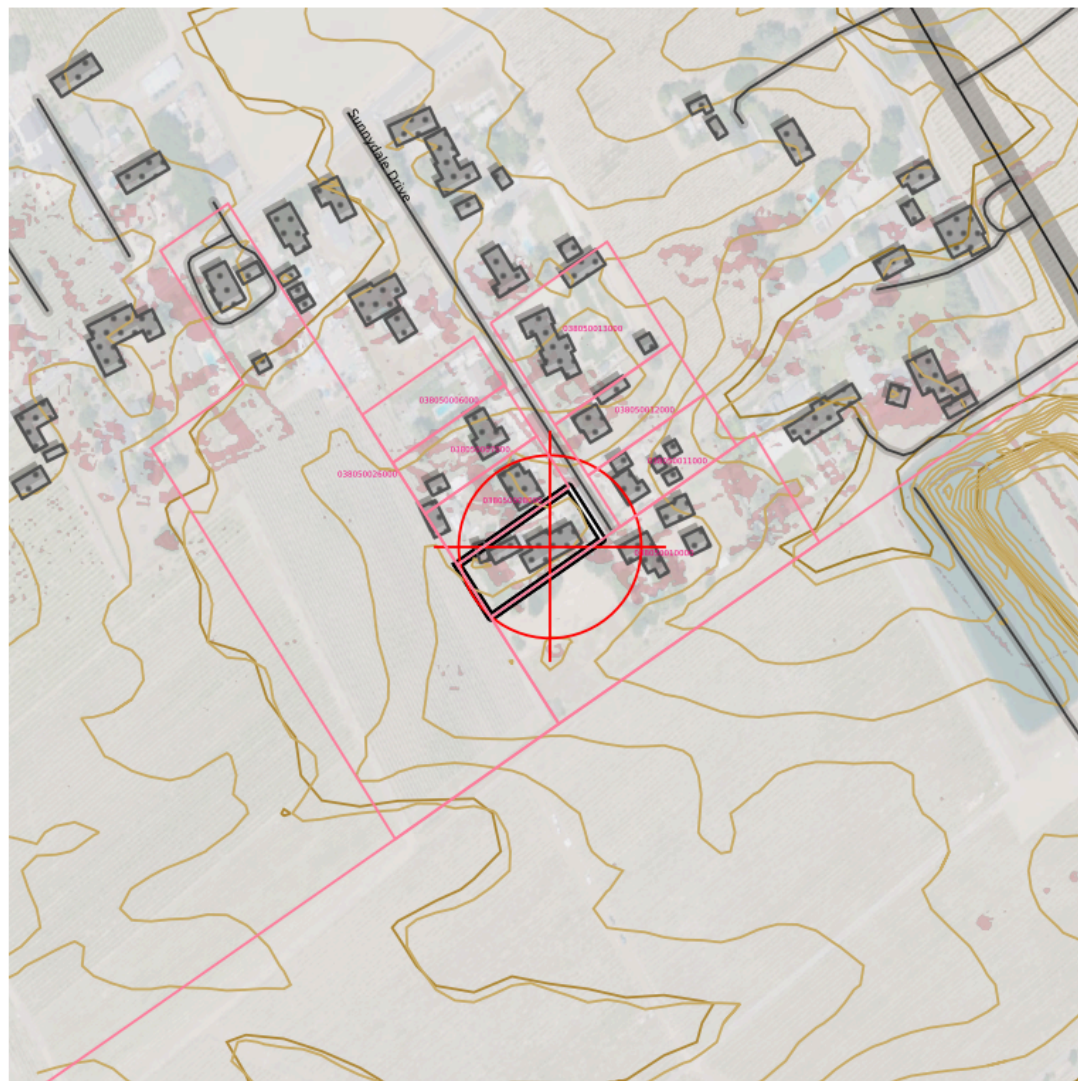


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 12, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – <https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate>

Current fire locations, smoke plumes, PM2.5 readings by location

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/GEOID18> and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

